WORKSHEET · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

Lesson 7: Randomness and Probability

Module 3 | 45 minutes | Random simulation | Grades 6–8 · ages 11–14

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. simulation ___ A. More trials → results closer to theory.
b. trial ___ B. How likely something is (0 to 1).
c. probability ___ C. One run of an experiment.
d. law of large numbers ___ D. A program that imitates a real process.

2. Plan your code

Before you build it: Coin-flip lab: set heads to 0. Repeat 100 times: if pick random 1 to 2 = 1, change heads by 1. Say the number of heads. Write or draw the blocks you will use.

3. Small samples

You flip a fair coin 10 times and get 7 heads. What's the best conclusion? Circle one.

a. A. That can happen by chance; try many more flips
b. B. The coin is definitely unfair
c. C. Random numbers are broken

4. Plan your code

Before you build it: Random walk: a Bug starts in the middle with the pen down. Repeat 200 times: turn right by pick random -90 to 90 degrees, move 6 steps, and bounce off edges. Watch the path it draws. Write or draw the blocks you will use.

5. Explain your thinking

Run your coin-flip lab five times. Record the results and explain why they’re different each time, and what would happen with 10,000 flips.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

- ☐ I can simulate a random event many times.
- ☐ I can compare results with theory.
- ☐ I can explain the law of large numbers.